Figure A.3 FSM for counting exact change (25 cents)

The starting state is S0, and the ending state of the FSM design diagram is S25. That is, beginning at the value 0, using 5C and 10C coins, you would like to reach the S25 state. The states possible are S5, S10, S15, and S20. You can see the transitions brought about by the customer inserting 5C and 10C coins, ending up with S25. The FSM in figure A.3 exhaustively enumerates all the logical possibilities. Assume that the customer is aware of the requirement to insert exact change.

You can draw this diagram with any tool of your choice. The version here was created with the widgets provided by the draw.io tool; draw.io does not explicitly provide an FSM diagram, but you can use the circles and arrows from the general template.

A.3 Structural diagrams

This category of UML diagrams helps you define the static structural design of your solution. We will study only one of them: the class diagram. You will learn to represent multiple classes and their relationships, using standard notations. Class diagrams are useful for defining the overall structure of a module or a smart contract solution.

A.3.1 Class diagrams